

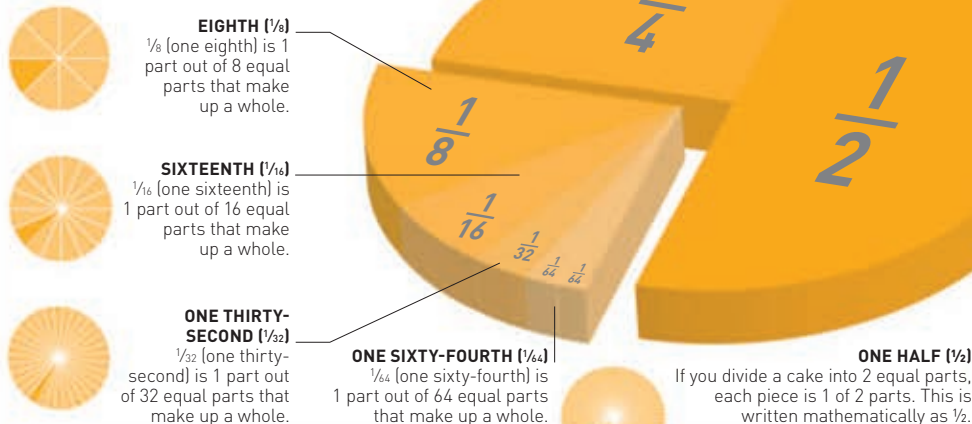
POSITIVE AND NEGATIVE NUMBERS

Positive numbers count up from zero; negative numbers count down from zero. This means they are less than zero. If you had £5 in your bank account and withdrew £10 from a cash machine, your bank balance would show as -£5.



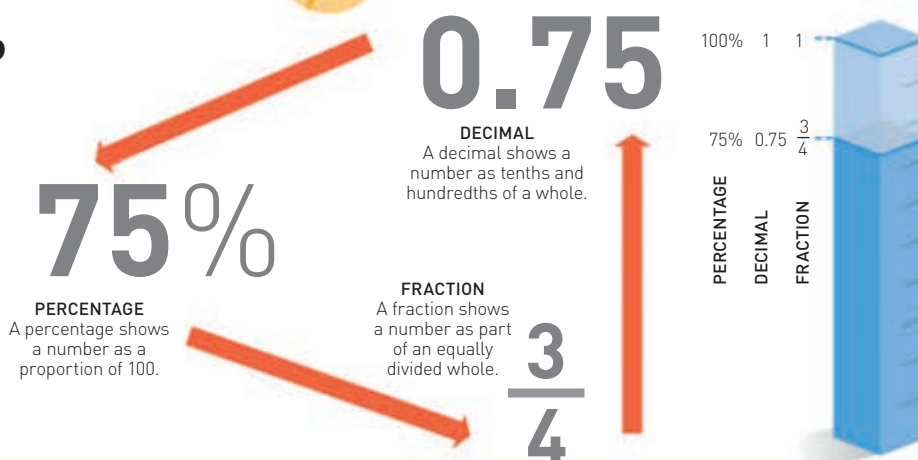
FRACTIONS

Fractions are a way of expressing parts of an object or number. If you cut a cake, for instance, into 2 equal parts, each piece is now 1 of 2 parts; this is written as 1 over 2, like this: $\frac{1}{2}$.



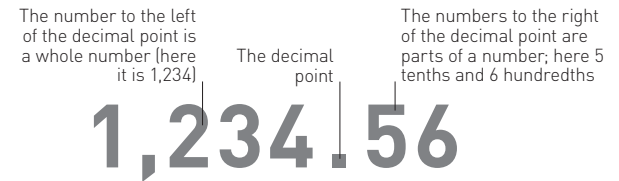
DECIMALS, FRACTIONS, AND PERCENTAGES

These are all ways of talking about parts of a number, or something that is less than a whole (such as half a cake, 50% of a class, or 0.5 of a metre). We can "translate" fractions, decimals, or percentages into each other. For instance, $\frac{3}{4}$ is the same as 75% or 0.75.



DECIMALS

Decimals are a way of expressing parts of things or numbers as tenths or hundredths of a whole number.



PERCENTAGES

Percentages are another way of talking about parts of an object or number. Here, the whole (such as the whole of a school class) is said to be 100 per cent, or 100%. Half the class is therefore half that: 50%. The whole can be broken into very fine parts up to 100%.

100% In a class of 100 children, 100% = 100 children

50% In a class of 100 children, 50% = 50 children

1% In a class of 100 children, 1% = 1 child

DECIMALS, FRACTIONS, AND PERCENTAGES ARE DIFFERENT WAYS OF SAYING THE SAME THING

COMMON NUMBERS

The table below shows some commonly used fractions, decimals, and percentages.

Decimal	Fraction	%	Decimal	Fraction	%
0.1	$\frac{1}{10}$	10%	0.625	$\frac{5}{8}$	62.5%
0.125	$\frac{1}{8}$	12.5%	0.666	$\frac{2}{3}$	66.7%
0.25	$\frac{1}{4}$	25%	0.7	$\frac{7}{10}$	70%
0.333	$\frac{1}{3}$	33.3%	0.75	$\frac{3}{4}$	75%
0.4	$\frac{2}{5}$	40%	0.8	$\frac{4}{5}$	80%
0.5	$\frac{1}{2}$	50%	1	1	100%

PRIME NUMBERS

These are special numbers that cannot be divided by any other number except themselves and 1. For example, 13 cannot be divided by any number other than 13 or 1. Numbers that can be divided by others are known as "composite numbers".

KEY TO TABLE

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PRIME NUMBER

A green box on the table indicates that the number is a prime number.

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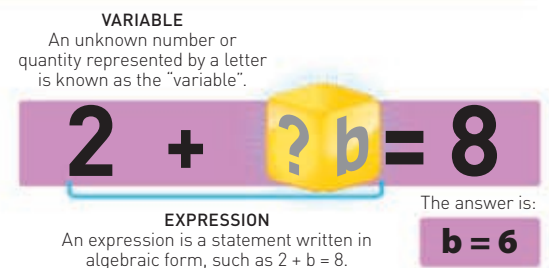
COMPOSITE NUMBER

A blue box indicates that a number is a composite number. The numbers it is divisible by are given as smaller numbers below it (2, 3, 7 in the example above).

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

ALGEBRA

When mathematicians are trying to work out a missing number in an equation, they use a symbol to represent the missing number. In this example, we know that 2 plus something (here called "b") equals 8.



AVERAGES

An average is the middle value of a set of data. The most common type of average is the mean, which is found by adding up a set of numbers then dividing the total by the amount of numbers in the set.

